

Lab 4 Tasks

Advanced Software Development Workshop
Spring 2026

Main Tasks

1. Teacher review feedback from last week
 - A. SRS partial documents
 - B. Environment diagram (context model) and use case diagram
2. Teacher explain Section 3 and 4.1
Updated version of Long Project Description, v1.1
3. Students
 - A. Update SRS partial documents according to feedback.
 - B. Create state transition diagrams and user interfaces for the specified cases.
 - C. Create a scenario for each of the specified cases.

Feedback on Context Model

- All components should be chosen from the given candidate list in the task ppt.
- Only login, MCS and BNBU user database should be included in the diagram.
- Some groups did not specify the boundary in the diagram.

Feedback on Use Case Diagram

➤ Completeness

- Most groups did well
- Some missed login
- Several groups missed supporting staff
- Some groups missed some use cases

Feedback on Use Case Diagram

➤ Consistency

- Most groups did well
- Some incorrect extensions
- Some incorrect inheritance between use cases:
 - Add, Delete and Modify cannot inherit Edit or Manage members,
- Login should be included by all use cases
- Nouns should not be use cases

Feedback on Use Case Diagram

➤ Redundancy

- Communicate use case include receiving message and making appointments
- Delete, Add, Modify are contained in the use case Edit
- Edit and Search include View
- Some groups gave too much detailed in operations; use cases should be main goals

Feedback on Use Case Diagram

➤ Notation

- Most groups did well
- Some problems:
 - Font size is too small
 - Extend notation has wrong direction
 - If the actors can clearly connect to the use cases without crossing any lines, then the actors and use cases can be in one diagram; otherwise more than one diagrams should be used.

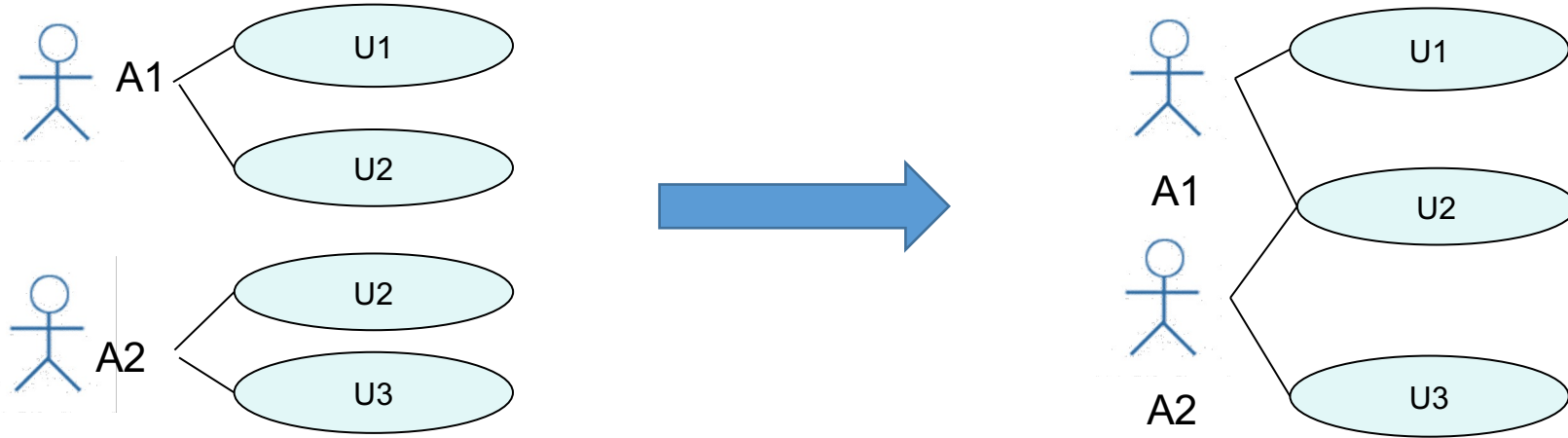
Feedback on Use Case Diagram

➤ Notation

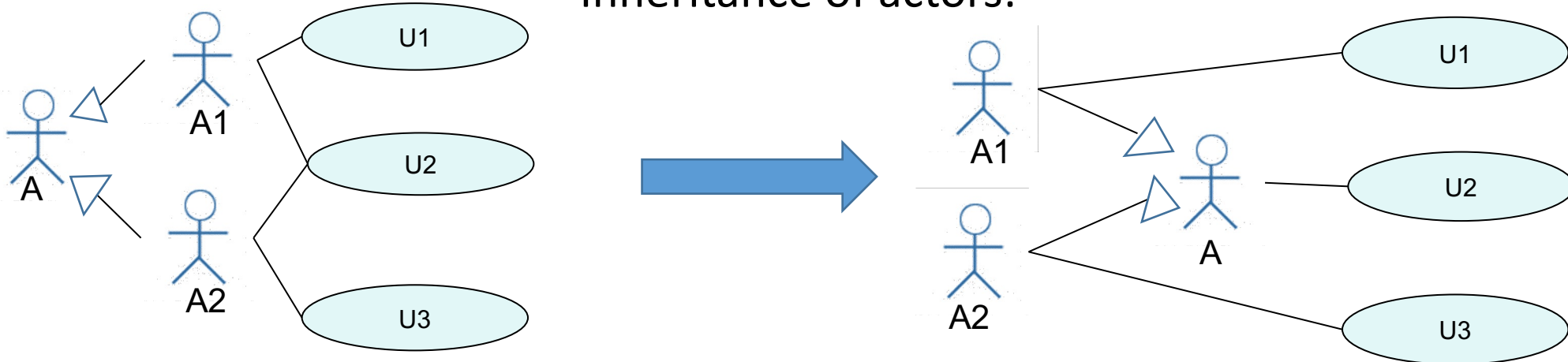
- Some problems (continue):
 - Crossed lines reduce readability
 - Dashed rectangle can be used to simplify the drawing when multiple use cases have same relationships with a use case.

Feedback

- Use Case Diagram: left side not good, and right side is better.
 - Same use case for different actors:



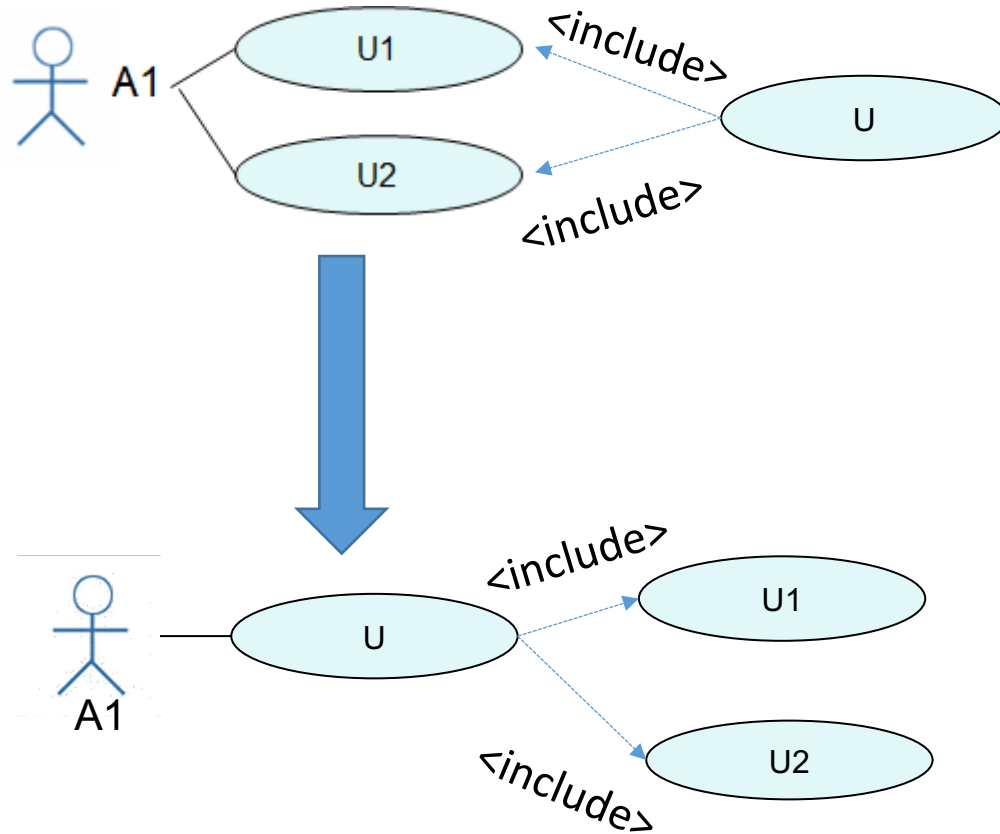
- Inheritance of actors:



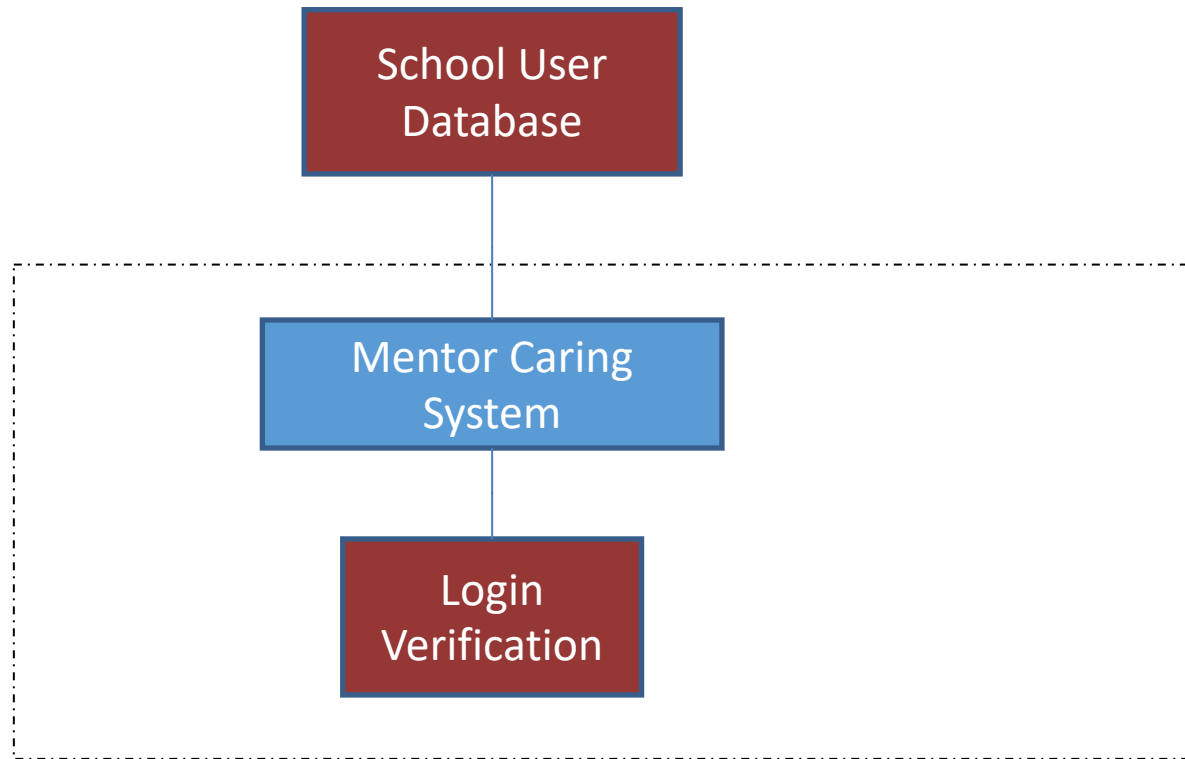
If there is no common use case for A, then A should not exist.

Feedback

- Use Case Diagram
 - Include use cases



Context model



Use case diagram

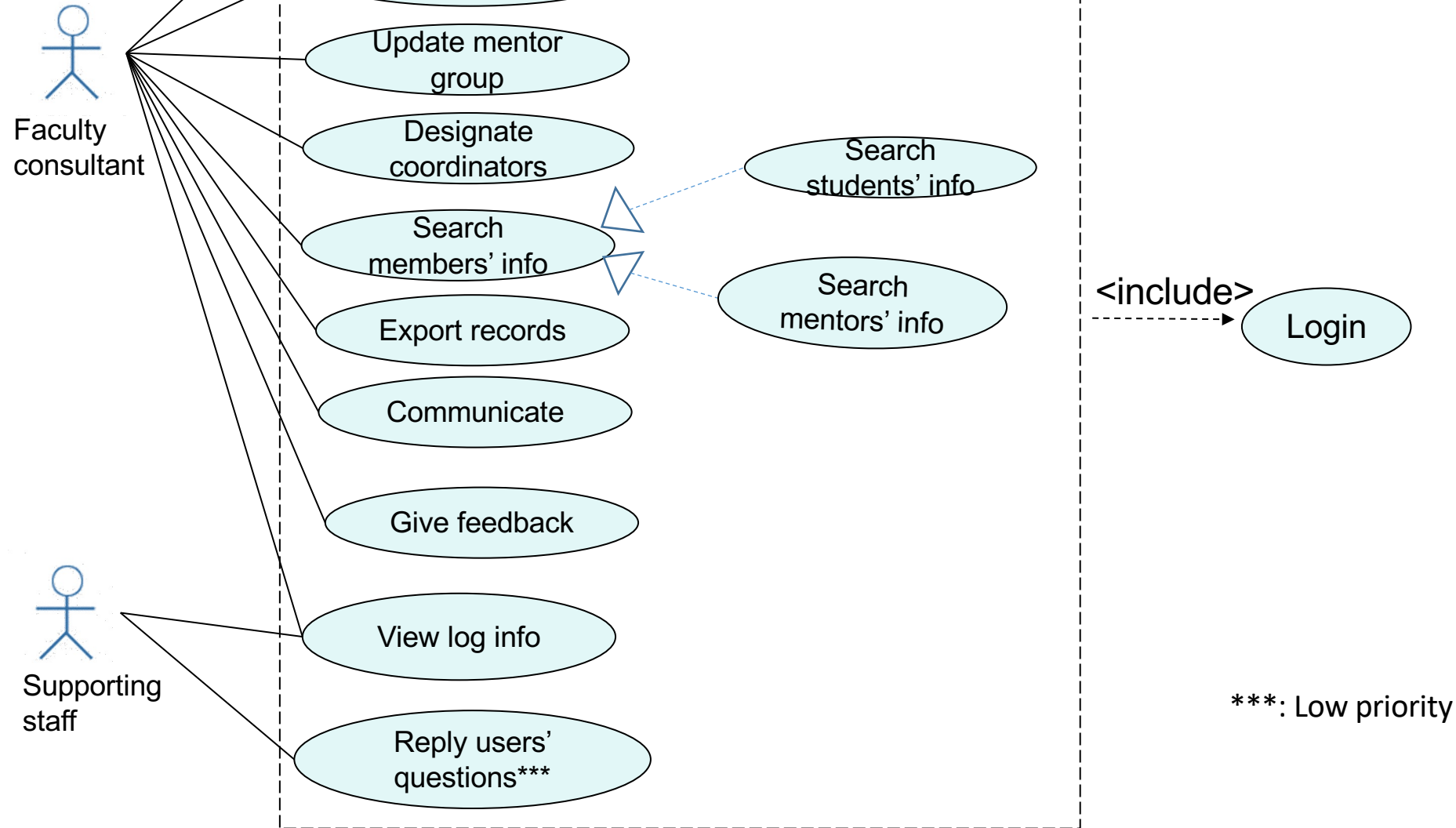


Figure 1

Use case diagram

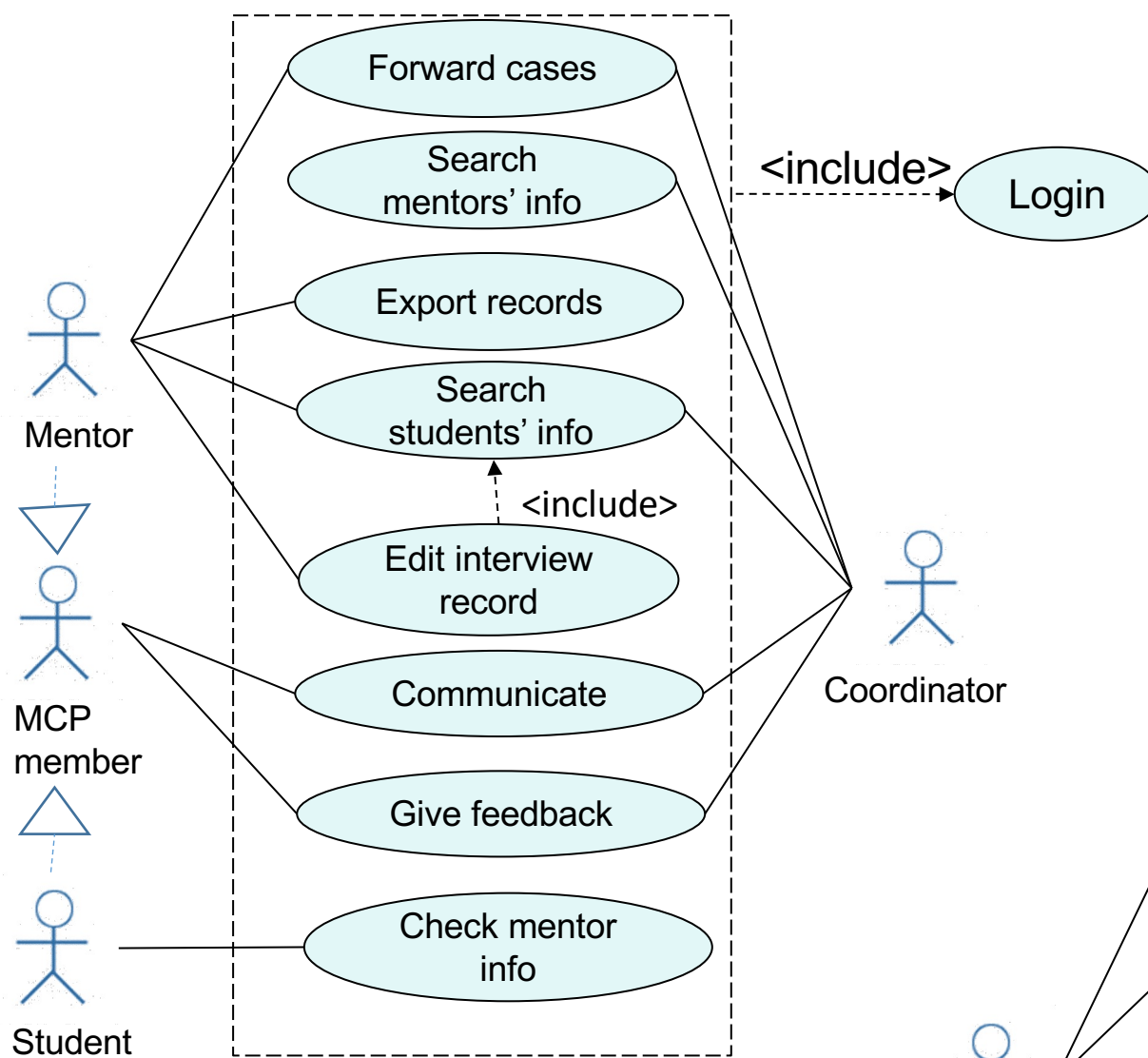


Figure 2

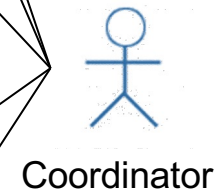
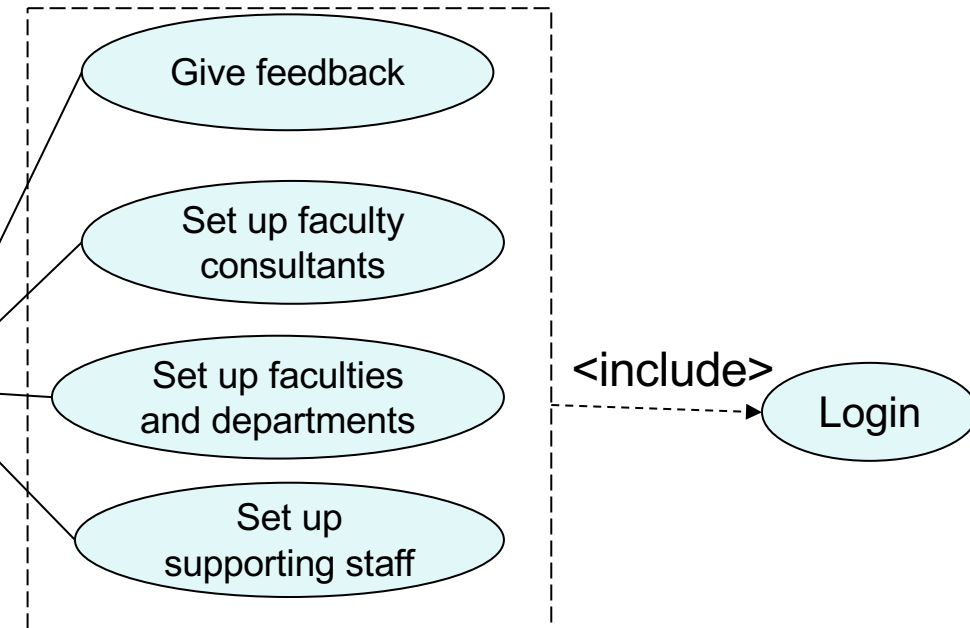


Figure 3



Administrator



Task2: Explain SRS Template

- **Teacher explains SRS Sections 3 and 4.1**
- For each **use case** for an **actor**, there must be a **corresponding state transition** diagram in a subsection in Section 3
 - Section **3.1** contains the state transition diagram for Use Case **1**, Section **3.2** contains the state transition diagram for Use Case **2**, etc.
 - If a use case is connected to two different actor roles, you can either
 1. Draw two different state transition diagrams, or
 2. Draw one state transition diagram if the difference can be clearly seen in the diagram.
- iSpace has updated version of **Long Project Description v1.1**.
 - Updated parts are **highlighted in yellow**.

Task 3A: Update SRS partial documents

- **Task 3A: Update** the SRS partial documents according to feedback
 - Correct the text part according to the feedback given in this PPT and by the grader.
 - Update (or replace) the context model and use case diagram with the solution we gave you in the earlier slides.
 - Unless your diagrams are near perfect, we suggest you use the diagrams we give you. Otherwise, you will continue to have more points deducted in future labs for keep making the same errors.

Task 3B and 3C: Students tasks

- **Task 3B:** Draw **state transition diagrams** and **user interfaces (UI)** for the specified use cases, plus the include use cases.
- **Task 3C:** Do the basic **scenario** for specified use cases, plus the included use cases.
- **This week**, you draw the diagrams for the **specified use cases** only; **next week** you will draw the diagrams for **all** the **use cases**.

Task 3B: State Diagrams and User Interfaces

- Section 3.1
 - Actor: Mentor, Coordinator
 - Use case: Search Student's Info
 - Description: Search for student information according to student ID. The interview record and other info of this student are displayed.
- Remarks:
 - 3.1 is included in 3.2 Edit Interview Record. (Must search for the student's info before editing it.) Combine the transition diagrams for 3.1 in 3.2 and put the combined diagram in 3.2.2. However, distinguish the actor as Mentor or Coordinator in the combined diagram. Coordinator can view the Student's info, but cannot edit it. Mentor can do both.
 - In 3.1.2, put the remark, "Transition diagram is included in 3.2.2."

Task 3B: State Diagrams and User Interfaces

- Section 3.2
 - Actor: Mentor
 - Use case: **Edit Interview Record**
 - Description: Search Student Info must be performed first. (See 3.1). The mentor can click Edit to add, modify or delete the interview record. If the record exists, the mentor can be modified or deleted it. If the record is empty, the mentor can enter a new record. The mentor can then Save or Cancel the changes.
- Remarks:
 - Combine the transition diagram of 3.1 Search Student's Info with the transition diagram for this use case. However, make a distinction whether the user is Mentor or Coordinator.
 - Put the **combined transition diagram** in **3.2.2**; add the comment, "Combined transition diagram for Search Student's Info (3.1) and Edit Interview Record (3.2)."

Task 3B: State Diagrams and User Interfaces

- Section 3.2
 - Actor: Mentor
 - Use case: **Edit Interview Record**
 - Description: The mentor is in a student's info page. (See 3.1, Search Student Info.) The mentor can click Edit to add, modify or delete the interview record. If the record exists, the mentor can be modified or deleted it. If the record is empty, the mentor can enter a new record. The mentor can then Save or Cancel the changes.
 - Combine the transition diagram of 3.1 (Search Student's Info) for Mentor with 3.2 (Edit Interview Record). Put the combined transition diagram in Section 3.2.2. Make a note this is the combined transition diagram of 3.1 and 3.2 for Mentor.

Task 3B: State Diagrams and User Interfaces

- Section 3.3
 - Actor: Faculty Consultant, Coordinator
 - Use case: **Search Mentor Info**
 - Description: Search for a mentor info, given a mentor's name, email address or group ID. The search result includes the mentor's name and ID of the group for which the mentor cares. The member list is displayed by clicking on the group ID. The record of a student is shown by selecting the name of the student in the member list.
 - Put this transition diagram in Section 3.3.2
 - Remark: you did the scenario for this use case in last week's lab.

Task 3B: State Diagrams and User Interfaces

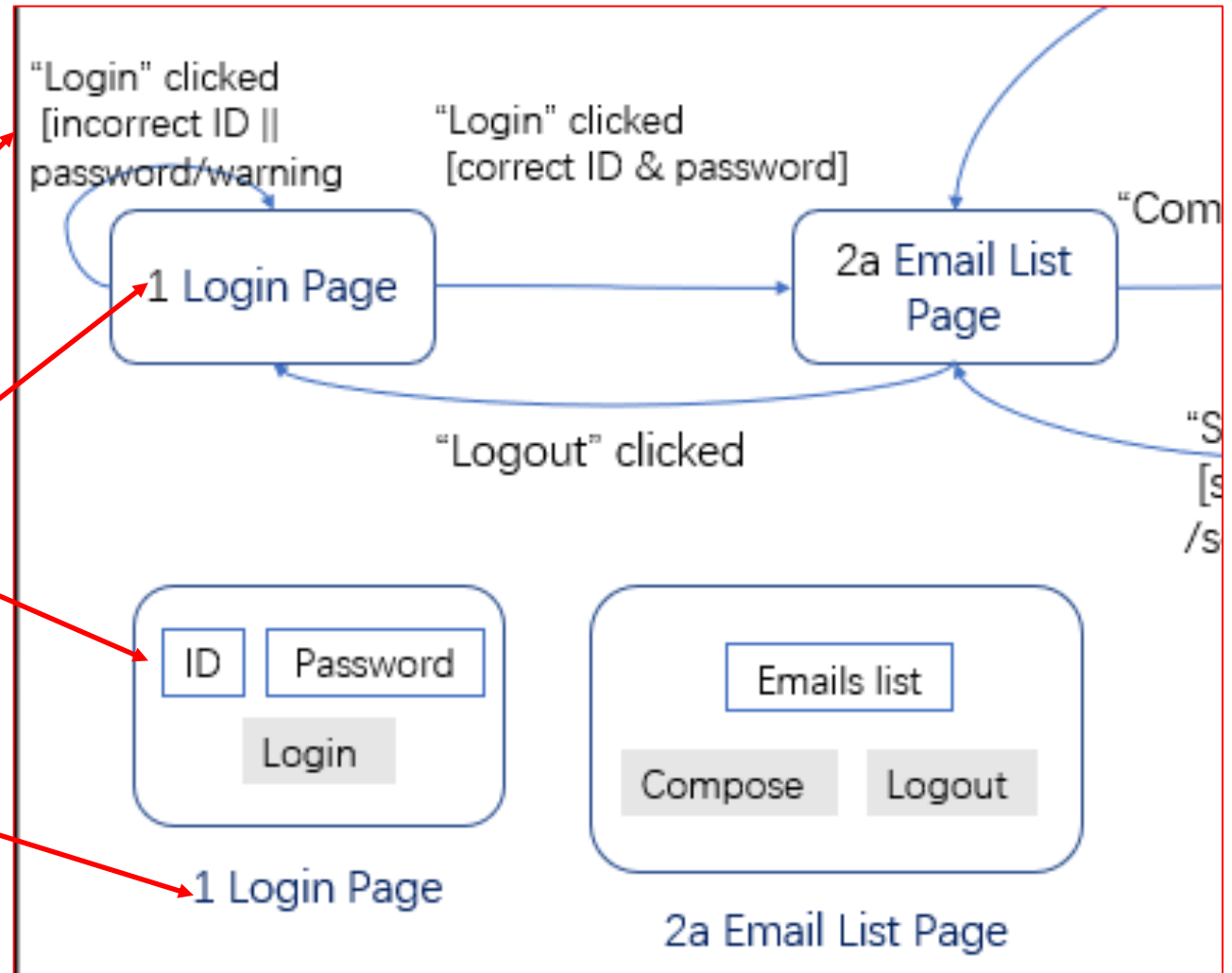
- Contrary to what is stated in the SRS template, put the **state transition diagram** in **Section 3.X.2**, not in **Appendix B**.
- User interface (UI)
 - For **each state** in the **state transition diagram**, there must be a **corresponding UI** in the **UI diagram**.
 - Please put the **UI diagram** in a **separate document** and mention that **document's name** in **Section 4.1**.
 - The grader needs to open the UI diagram and the transition diagram simultaneously.
 - Each UI should have a number and label that are meaningful for referencing. The **number** and **label** for the **UI must exactly match** those in the **corresponding state** in the **state transition diagram**.

Example from Lecture 4 class exercise

Each state in the transition diagram must **exactly match** the corresponding UI.

All controls (ID, Password and Login button) match

Number and label exactly match



Task 3C: Basic Scenario

- **Task 3C:** Do the basic **scenario** for the specified use cases.
 - Please state the assumption of each scenario, i.e. what had already happened before the scenario.
 - You should **do** the **scenario before** the **transition diagram** because the scenario helps you to figure out the sequence of events for the transition diagram.
- Put the scenario under the state transition diagram in Section 3.X.2.
- For 3.X.3 functional requirements,
 - Usually, the state transition diagrams and UI contain enough information.
 - So in 3.X.3, enter only the functional requirements that are **not described in** the **transitional diagrams** and **UI**.

Submission

- Each group submits the partial SRS (**two documents, ONE copy**) to iSpace and GitLab
 - Deadline*: **23:55 pm, Fri 20 March 2026.**
 - On the cover page next to the student's name, please indicate the contribution of each student: "FULL", "FAIR", "LITTLE" or "NONE".

* There is a few minutes of grace period after the DDL in case any network or unusual problems occur. But do respect the DDL because the length of the grace period is not guaranteed.